

MALLEPALLI MAHESH

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EDUCATION

SHADAN COLLEGE OF ENGINEERING AND TECHNOLOGY

Hyderabad, TG

Bachelors of Technology in AIML - 7.2

2022-2026

EXPERIENCE

- GeeksforGeeks - Campus Mantri(Campus Ambassador)** Jan 2026 - July 2026
Represented GeeksforGeeks on campus, promoting technical resources and initiatives. Organized events and workshops to enhance student engagement and coding culture.

PROJECTS

Dental Voice Assistant – Conversational AI Voice Assistant for Dental Support: [GithubLink](#)

- Built a **real-time AI voice assistant** using **Twilio**, **Deepgram**, and **OpenAI**, enabling end-to-end voice interactions for dental assistance over phone calls
- Implemented LLM function calling for **symptom analysis**, **dentist recommendations**, and **appointment booking**, powered by an asynchronous **Python WebSocket** backend for low-latency processing.
- MongoDB** integration for data persistence and a **config-driven architecture** to manage audio settings, agent behavior, and API workflows.

Transaction Assistant- AI powered Q&A Platform: [GithubLink](#) [YoutubeLink](#)

- Built an AI-powered Q&A platform enabling **NLP** queries on transaction history using **RAG** technology, **FAISS** vector database, and **HuggingFace** embeddings.
- Integrated **FastAPI** for efficient backend processing and **LangChain** pipeline powered by **Groq LLM** to deliver accurate, context-aware answers instantly.
- Developed a responsive **React**-based frontend with a floating chatbot interface for real-time transactions including UPI and payment records.

Chat with SQL Databases - Ask questions about your database:[GithubLink](#) [YoutubeLink](#)

- Designed a natural language interface to query **SQL** databases using **LLMs** and **LangChain** agents, automatically converting plain English into optimized SQL queries.
- Integrated **LangChain SQL Agent** framework with **Streamlit** to provide a user-friendly interface, enabling seamless database exploration without requiring SQL expertise.
- Implemented support for multiple database backends including **MYSQL** and **SQLiteDB**, with real-time response generation and human-readable answers from structured tables.

SKILLS

Programming Languages	- Python Programming
AI Frameworks	- Langchain, LangGraph, LangSmith, Transformers, NLP, LLM's, RAG, NLTK, Streamlit, Pandas
Vector Database	- FAISS, Qdrant, Pinecone
Data Storage	- MYSQL, MongoDB
IDEs	- Visual Studio, Jupyter, Google Colab
Soft Skills	- Communication, Networking, Spearheaded, Adaptability

CERTIFICATIONS

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|--|---------------------|
| • Generative AI With Langchain and Huggingface | - Udemy |
| • Claude in Amazon Bedrock | - Anthropic |
| • Tata-GenAI Powered Data Analytics Job Simulation | - Forage |
| • LangGraph Ambient Agents | - LangChain Academy |
| • Python UpSkill | - GeeksforGeeks |

CO-CURRICULAR ACTIVITIES

- Represented the college team in **District-level cricket**, contributing to multiple victories through consistent performance as a key player. Demonstrated discipline, teamwork, and strategic decision-making under pressure for the year **2025–2026**.